Interaction and Concurrency

Individual Assignment on Reactive Systems

Until next 30 May 2025, please provide a complete, individual answer and quote suitably any reference (paper, book, software) used.

Problem 1

A relation R over the state space of a labelled transition system is a *word bisimulation* if, whenever $\langle p, q\rangle \in R$ and $s \in N^*$, we have

$$p \xrightarrow{s} p' \ \Rightarrow\ \exists_{q' \in S_2}.\ q \xrightarrow{s} q' \wedge \langle p', q'\rangle \in R$$
$$q \xrightarrow{s} q' \ \Rightarrow\ \exists_{p' \in S_1}.\ p \xrightarrow{s} p' \wedge \langle p', q'\rangle \in R$$

1. Define formally relation $\xrightarrow{s}$, for $s \in N^*$
2. Two states are *word bisimilar* iff they belong to a word bisimulation. Discuss formally whether two states p and q are word bisimilar iff $p \sim q$.

Problem 2

Make yourself familiar with mCRL2 through its webpage. Then, consider the following specification of a control system for a crossing between a road and a railway. Events car and train modelled, respectively, a car or a train approaching the cross. Actions up e dw stand for the opening and closing of the protection bar to prevent cars to cross. Similarly, green and red model the semaphore for trains. Finally, events $\overline{\mathsf{ccross}}$ and $\overline{\mathsf{tcross}}$ come from sensors which register the actual cross of a car or a train, respectively.

$$\begin{aligned}
\mathsf{Road} &\ \widehat{=}\ \mathsf{car}.\mathsf{up}.\overline{\mathsf{ccross}}.\overline{\mathsf{dw}}.\mathsf{Road}\\
\mathsf{Rail} &\ \widehat{=}\ \mathsf{train}.\mathsf{green}.\overline{\mathsf{tcross}}.\overline{\mathsf{red}}.\mathsf{Rail}\\
\mathsf{Signal} &\ \widehat{=}\ \overline{\mathsf{green}}.\mathsf{red}.\mathsf{Signal} + \overline{\mathsf{up}}.\mathsf{dw}.\mathsf{Signal}
\end{aligned}$$

$$\mathsf{C}\ \widehat{=}\ (\mathsf{Road} \mid \mathsf{Rail} \mid \mathsf{Signal})\backslash_{\{\mathsf{green},\mathsf{red},\mathsf{up},\mathsf{dw}\}}$$

1. Explain the behaviour of this process and sketch its synchronisation diagram.
2. Use mCRL2 to draw the transition system of process C. What more can you do with mCRL2 to analyse the behaviour of process C?